Zhe Xue

(+86) 13896366299 · xuezhe.xz@gmail.com · xuezhe-zachary.github.io

Education

Sichuan University, College of Computer Science, China Sep. 2024 – Jun. 2027 (Expected)
M.Eng. in Software Engineering **Overall Average: 89.3/100**

Sichuan University, School of Software Engineering, China Sep. 2020 – Jun. 2024
B.Eng. in Software Engineering **Overall Average: 86.5/100**

Publications

1. **Z. Xue**, C. Sun, W. Zheng, J. Lv, and X. Liu. TargetSA: adaptive simulated annealing for target-specific drug design. *Bioinformatics*, 41(1), btae730, 2025. (JCR Q1, IF = 5.5)
2. X. Liu†, **Z. Xue**†, M. Luo, B. Ke, and J. Lv. Anesthetic drug discovery with computer-aided drug design and machine learning. *Anesthesiology and Perioperative Science*, 2, 7, 2024. (JCR Q1, IF = 7.5) † Equal contribution; Z. Xue is the first student author.
3. **Z. Xue**†, W. Ai†, J. Sun, et al. A multi-agent molecular optimization framework leads to a rapid-recovery intravenous anesthetic candidate with an improved safety margin. *Preprint*, 2026. † Equal contribution.
4. Y. Li, X. Zeng, **Z. Xue**, P. Zeng, Z. Zhang, and Y. Wang. Incorporating the Refractory Period into Spiking Neural Networks through Spike-Triggered Threshold Dynamics. *Proceedings of the 33rd ACM International Conference on Multimedia*, 10876–10885, 2025.

Research Experience

AI for Drug Discovery Jun. 2022 – Present
Data Intelligence and Computational Art Laboratory, Sichuan University; Department of Anesthesiology, Laboratory of Anesthesia and Critical Care Medicine, National-Local Joint Engineering Research Centre of Translational Medicine of Anesthesiology, West China Hospital, Sichuan University

TargetSA: Adaptive Simulated Annealing for Target-Specific Drug Design 2024

- Addressed the challenge of navigating vast discrete chemical space for target-specific ligands under competing affinity and drug-property objectives.
- Designed a history-guided GNN to identify promising editing sites and combined four graph-edit operations with reversible simulated annealing to re-accept temporarily suboptimal candidates and escape local optima.
- Achieved a mean Vina score of -9.09 across 100 CrossDocked2020 pockets, with 79.4% of generated molecules outperforming the corresponding reference ligands.
- Published as first author in *Bioinformatics* and released an open-source implementation.

MASCOT: Multi-Agent Molecular Optimization for Anesthetic Discovery 2024–2026

- Reframed multi-objective molecular optimization as adaptive search control, enabling the search to respond as property bottlenecks shift over time.
- Designed and implemented MASCOT, a controller–executor framework in which three specialized LLM agents adapt objective priorities, exploration versus refinement, and trajectory memory, while a chemically constrained graph editor performs valid and auditable molecular modifications.
- Achieved the best overall optimization performance across four multi-objective generation benchmarks and two similarity-constrained SARS-CoV-2 Mpro lead-optimization settings.
- Applied MASCOT to marketed anesthetic remimazolam (RM), generating 2,729 candidates and prioritizing RM-1 for synthesis. Subsequent side-chain derivatization and experimental screening advanced RM-7 as a rapid-recovery intravenous anesthetic candidate, with a 61% lower ED_{50}, 62% shorter time to post-anesthesia walking recovery, and a $3.7\times$ higher therapeutic index than RM.
- Co-first-author preprint with an open-source implementation.

Flexible-Pocket 3D Molecular Generation for Structure-Based Drug Design 2025

- Explored flexible-pocket 3D molecular generation beyond the common practice of conditioning ligand generation on a fixed protein structure.

- Designed an E(3)-equivariant diffusion framework jointly modeling ligand generation and pocket conformational refinement through residue transformations, side-chain torsions, and protein–ligand interactions.
- Built an end-to-end prototype and characterized key challenges in flexible protein–ligand generation, including limited apo-to-holo supervision, coupled sampling errors, and training–inference mismatch.

Spectral Graph Diffusion for Molecular Generation 2026

- Explored Laplacian spectra as compact representations of global connectivity and ring topology for diffusion-based molecular graph generation.
- Implemented and evaluated spectral-domain diffusion, Transformer-based graph score models, and spectral guidance on QM9 and ZINC250K, with explicit modeling of bond types and ring structures.
- Identified that spectral features capture global topology but do not sufficiently preserve local chemical constraints, suggesting their use as auxiliary guidance rather than a standalone generation space.

Research Interests

Research Focus: Reliable generative AI and scientific agents for closed-loop molecular discovery.

- **Generative Molecular Design:** molecular generation and optimization; protein–ligand modeling
- **Scientific Agents:** planning, tool use, multi-agent coordination, and feedback-driven learning
- **Reliable AI for Science:** controllability, robustness, uncertainty-aware evaluation, and biological safety

Honors & Awards

Personal Honors

- [National] **National Scholarship** — awarded to the top 0.2% of university students nationwide. 2023
- [Provincial] Outstanding Graduate of Sichuan Province. 2024
- [University] Outstanding Graduate Student, Sichuan University. 2025
- [University] Top Ten Students Nominee — 15 nominees for the university's highest student honor. 2024
- [University] Scholarship for "Star of Reading" — 10 recipients university-wide. 2024
- [University] Scholarship for Gratitude to Modern Chinese Scientists — 12 recipients university-wide. 2023
- [School] Scholarship for "Software Elite" — 3 recipients among 213 students. 2024

Competitions & Innovation Programs

- [National] First Prize, 25th China Robotics and Artificial Intelligence Competition — team leader. 2023
- [National/Provincial] Undergraduate Innovation and Entrepreneurship Training Program — national-level core researcher (2023); provincial-level project leader (2023) and team member (2022). 2022–2023
- [Provincial] Second Prize, 16th China Chengdu International Software Design and Application Competition — team leader. 2022

Extracurricular

National University of Singapore, School of Computing, Singapore May 2022 – Jul. 2022

Summer Workshop; Third Prize in the Web Mining course **Grade: A**

Skills & Service

- **Programming:** Python, PyTorch, RDKit, Linux
- **Languages:** TOEFL 104 (2022), CET-4 615, CET-6 570
- **Leadership & Service:** Class President (2020–2024); Student Affairs Assistant (2022–2024); Undergraduate Peer Mentor (2024–2025); Teaching Assistant (2024, 2026)